

1(a)

$$\left(\frac{\langle t_1 \rangle - t}{t} \right) \times 100\% = \left(\frac{0.417 - 0.452}{0.452} \right) \times 100\% = -7.82\%$$

$$\left(\frac{\langle t_2 \rangle - t}{t} \right) \times 100\% = \left(\frac{0.460 - 0.452}{0.452} \right) \times 100\% = 1.77\%$$

Accuracy: closeness of measurement the true value

percentage difference of $\langle t_2 \rangle$ from accepted value less than that of $\langle t_1 \rangle$,
group 2 more accurate

Range of measurement for group 1 = $0.440 - 0.390 = 0.050$ s

Range of measurement for group 2 = $0.570 - 0.395 = 0.175$ s

Precision: how close repeated measurements are to each other

range of measurement for group 1 less than that for group 2,
the measurement for group 1 is more precise.

1(b)(i)

$$\frac{\Delta E}{E} = \left(\frac{\Delta L}{L} + \frac{\Delta(F/e)}{(F/e)} + 2 \frac{\Delta D}{D} \right)$$

$$\Delta E = \left(\frac{0.01}{2.00} + \frac{5}{95} + 2 \left(\frac{0.05}{1.15} \right) \right) (1.82923 \times 10^{11})$$

$$= 0.264 \times 10^{11} \text{ N m}^{-1}$$

1(b)(ii)

$$E = (1.8 \pm 0.3) \times 10^{11} \text{ N m}^{-1}$$